

005 · Amenities — Functions Calling Functions — English Worksheet (Intermediate)

Topic: Amenities — Functions Calling Functions · **Course:** Theme Park Creator · **Time:** about 45 minutes

Your park has stalls. Now it needs **amenities** — a **bathroom**, a **janitor room**.

Both are rooms. So you write **one** `build_room` helper. Both amenity functions **call** it, then add their own fitting.

These are the blocks you use this lesson:

```
blocks.fill(QUARTZ_BLOCK, pos(0, 0, 0), pos(4, 3, 4), FillOperation.HOLLOW)
blocks.fill(AIR, pos(2, 1, 0), pos(2, 2, 0), FillOperation.REPLACE)
blocks.place(CAULDRON, pos(1, 1, 1))
```

`blocks.fill(material, corner1, corner2, ...)` fills a box between two corners.
`FillOperation.HOLLOW` = outer **walls** only — the middle is **cleared to air**.
`FillOperation.REPLACE` = the **whole solid** box. `blocks.place(block, pos(x, y, z))` puts **one** block in **one** spot. `CAULDRON` = the toilet. `BARREL` = the janitor's supplies.

1 · Meet Helper Functions 🧰

A **helper function** does one job. Other functions **call** it instead of writing that job again.

```
build_bathroom(5)      build_janitor_room(4)
|                       |
├─► build_room(...)    ─► build_room(...) ← ONE helper
|                       |
└─► place CAULDRON     └─► place BARREL
```

Here is the helper. It builds walls, a floor, and a doorway:

```
def build_room(wall, floor, size):
    blocks.fill(wall, pos(0, 0, 0), pos(size, 3, size), FillOperation.HOLLOW)
    blocks.fill(floor, pos(0, 0, 0), pos(size, 0, size), FillOperation.REPLACE)
    blocks.fill(AIR, pos(2, 1, 0), pos(2, 2, 0), FillOperation.REPLACE)
```

Why is writing `build_room` once better than writing those three fills in both amenity functions?

2 · Predict 🧠

Read each function. Write what you think happens.

```
def build_bathroom(size):  
    build_room(QUARTZ_BLOCK, STONE, size)  
    blocks.place(CAULDRON, pos(1, 1, 1))  
  
build_bathroom(5)
```

What are the walls made of? What is the floor made of? How wide is the room?


```
def build_janitor_room(size):  
    build_room(OAK_PLANKS, STONE, size)  
    blocks.place(BARREL, pos(1, 1, 1))  
  
build_janitor_room(4)
```

Both amenities call the same helper. What is different about the two rooms?

3 · Spot the Bug 🐛

Each block is broken. Read what it should do, rewrite it so it works, then explain why the original was wrong.

Bug A — This should build a bathroom: a room, then a cauldron. But only a **lonely cauldron** appears, with no room around it.

```
def build_bathroom(size):  
    blocks.place(CAULDRON, pos(1, 1, 1))  
  
build_bathroom(5)
```

Hint: the helper exists, but nobody calls it.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — This should give the bathroom **quartz walls** and a **stone floor**. But the walls come out stone and the floor comes out quartz.

```
def build_bathroom(size):  
    build_room(STONE, QUARTZ_BLOCK, size)  
    blocks.place(CAULDRON, pos(1, 1, 1))  
  
build_bathroom(5)
```

Hint: the helper's parameters are `(wall, floor, size)`. Arguments land in those boxes **by position**.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — This should build a room with a cauldron inside. The room appears, but the **cauldron is gone**.

```
def build_bathroom(size):  
    blocks.place(CAULDRON, pos(1, 1, 1))  
    build_room(QUARTZ_BLOCK, STONE, size)  
  
build_bathroom(5)
```

Hint: `FillOperation.HOLLOW` clears the middle of the room to air. The cauldron is in the middle.

Write the fixed code:

Why was it wrong? Why does your fix work?

In your homework world, write **one** helper and **two** functions that call it:

```
def build_room(wall, floor, size):
def build_bathroom(size):
def build_janitor_room(size):
```

```
build_room builds:
```

- **walls** from the `wall` material with `FillOperation.HOLLOW`
- a **floor** from the `floor` material
- a **doorway** carved with `AIR` so a visitor can walk in

`build_bathroom` calls `build_room`, then places a `CAULDRON` inside. `build_janitor_room` calls `build_room` with different materials, then places a `BARREL` inside.

Neither amenity function has a `blocks.fill` of its own. The walls, floor and doorway all come from the helper.

Run `build_bathroom(5)` . Then **walk somewhere new** and run `build_janitor_room(4)` , so the two rooms stand apart.

Write your helper and your two amenity functions:

5 · Show Your Work 📷 🎥

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit ✓

Send this worksheet + your video to teacher on KakaoTalk.